of tissue on the roots which are called root nodules (Figure 11.3). Root nodules are fleshy light pink colored globose structures seen on the roots. The bacteria inside the root nodules transform into swollen, mishappened forms. These are called **bacteroids**. The bacteroids are capable of nitrogen fixation (Figure 11.4)

3. Reduction of N≡N to ammonia in root nodules

This biochemical process is catalysed by an enzyme called **Nitrogenase** present in bacteroids and happens under diminished O_2 levels. The O_2 levels in the nodules are controlled by an oxygen binding protein called **leghemoglobin**. This is a red, iron containing protein which can keep the nodule environment free of oxygen.

Nitrogenase

The enzyme nitrogenase is a complex enzyme consisting of 2 enzymes, dinitrogenase reductase and dinitrogenase. Electrons from organic compounds like pyruvate are passed on to dinitrogenase reductase first and then to dinitrogenase which in turn passes them to N≡N thus reducing it to NH_4. This reduction needs 16 ATP, ferredoxin and cytochromes.

$$N_2 + 8_e^- + 8_{H^+} \rightarrow 2NH_3 + H_2$$

$$N \equiv N + 4H \longrightarrow HN{=}NH + 2H \longrightarrow$$
$$H_2N\text{-}NH_2 + 2H \longrightarrow 2NH_3$$

HOTS

Will nitrogen fixation occur in the presence of air? What will be the fate of nitrogenase enzyme in aerobic condition?



Figure 11.3: Root nodules in the leguminous plant root

11.1.3 Phosphorus Cycle

The element phosphorus (P) is an essential macro element in all living organisms. They are found in nucleic acids and phosphate esters. It is an essential component of ATP and other high energy phosphates and phospholipids.

Reservoirs of Phosphorus

1. Phosphate rock like apatite, a large inert reservoir
2. Marine and aquatic sediments
3. Dissolved phosphates in soils and waters
4. Organic phosphates in dead and living organisms.

Phosphorous transformations mostly happen as inter conversion of inorganic to organic phosphate and insoluble form to soluble phosphates.

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